

Operations Research

Introduction, History, Applications & Limitations

"OR turns data into decisions." — Anonymous



Syllabus

Unit 1: Introduction to Operations Research 5 Hrs

Introduction, History of Operations Research, Stages of Development of Operations Research, Relationship between Manager and OR Specialist, OR Tools and Techniques, Applications of Operations Research, Limitations of Operations Research

Unit 2: Linear Programming Problem 10 Hrs

Introduction to Linear Programming, Linear Programming Problem Formulation, Formulation with Different Types of Constraints, Graphical Analysis of Linear Programming, Graphical Linear Programming Solution, Multiple Optimal Solutions, Unbounded Solution, Infeasible Solution, Basics of Simplex Method, Simplex Method Computation, Simplex Method with More Than Two Variables, Primal and Dual Problems, Economic Interpretation

Unit 3: Transportation and Assignment Problem 8 Hrs

Transportation Problems definition, linear form, Solution methods: North West corner method, least cost method, Vogel's approximation method. Degeneracy in transportation, Modified Distribution method, unbalanced problems and profit maximization problems. Transshipment Problems. Assignment Problem Structure and Solution: Short-Cut Method (Hungarian Method), Unbalanced Assignment Problem, Infeasible Assignment Problem, Maximization in an Assignment Problem, Crew Assignment Problem.

Unit 4: Queuing Theory 6 Hrs

Basis of Queuing theory, elements of queuing theory, Kendall's Notation, Operating characteristics of a queuing system, Classification of Queuing models.

Unit 5: Inventory Control 6 Hrs

Inventory classification, Different cost associated to Inventory, Economic order quantity, Inventory models with deterministic demands, ABC analysis.

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Unit 6: Replacement Theory 6 Hrs

hrs. Introduction, Replacement of capital equipment which depreciated with time, replacement by alternative equipment, Group and individual replacement policy

Unit 7: Game Theory 7 Hrs

Introduction, Characteristics of Game Theory, Two Person, Zero sum games, pure strategy. Dominance theory, Mixed strategies (2×2 , $m \times 2$), Algebraic and graphical methods

Inventory models

References/ Suggested Readings:

Hillier, F.S.& Lieberman, G.J. (1995). Introduction to Operations Research, 7th edition.

The McGraw-Hill Companies, Inc.

Natarajan, A. M.; Balasubramani, P. & Tamilarasi, A. (2007). Operations Research.

Pearson Education Inc.

Sharma, J.K. (2009). Operational Research: Theory and Application. Macmillan Publishers India Ltd.

Taha, H.A. (2017). Operations Research: A Introduction, 10th edition, Global edition, Pearson Education, Inc. Pearson Prentice Hall.

Wagner, H. N. (2003). Operations Research by, Prentice hall. N D Vohra, Tata McGraw Hill.

Winston, L.W. (2004). Operations Research: Applications and Algorithms, Indian University, 4th edition.

Evaluation

Examination Scheme				
Internal Assessment		External Assessment		Total
Theory	Practical	Theory	Practical	
40		60	-	100

1. Introduction to Operations Research

- Operations Research (OR) is the application of scientific and mathematical methods to study and analyze problems involving complex systems for decision-making, optimization, and efficiency.
- **Objective:** To assist in decision-making and improve or optimize system performance under given constraints.
- **Domains:** Mathematics, Computer Science, Economics, Engineering...
- **Examples**
 - Optimizing supply chain logistics to minimize transportation cost and delivery time.

Also used in energy, telecom, retail pricing, and sports analytics for data-driven decisions.

1. Introduction to Operation Research

- **Examples**

- Optimizing supply chain logistics to minimize transportation cost and delivery time.
- Healthcare & Hospitals – Staff scheduling, patient flow, and vaccine distribution using queuing and optimization.
- Manufacturing & Production – Job scheduling, assembly line balancing, and quality control for efficiency.
- Finance & Risk – Portfolio optimization, credit scoring, and algorithmic trading with predictive models.
- Public Services & Defense – Traffic management, disaster response, military logistics, and cybersecurity strategies.



- A scientific approach to decision-making that optimizes the performance of complex systems.
- **Key Objectives:** Efficiency, cost reduction, resource allocation.
- **Applications:** Military, logistics, healthcare, manufacturing, etc.



→ Early Origins (Pre-20th Century)

- ◆ Ancient Examples:
 - Babylonian mathematicians (optimization problems).
 - Roman military logistics (supply chain management).
- ◆ Industrial Revolution:
 - Frederick Taylor's "Scientific Management" (1911).
 - Henry Gantt's scheduling charts (Gantt charts).

→ World War II – The Birth of Modern OR

- ◆ Military Applications:
 - Radar deployment (UK, "Blackett's Circus").
 - Anti-submarine warfare (Allied forces).
 - Aircraft sortie optimization.
- ◆ Patrick Blackett (Nobel laureate, physics)
- ◆ George Dantzig (Linear Programming)



→ Post-War Expansion (1940s–1950s)

- ◆ Transition to civilian applications:
 - Manufacturing (Ford, Toyota).
 - Transportation (airline scheduling).
 - **Key Developments:**
 - Simplex Method (Dantzig, 1947).
 - Game Theory (John von Neumann).



→ Growth in Academia and Industry (1960s–1980s)

- **OR in Academia:**
 - Universities establish OR departments.
 - Journals (*Operations Research*, *Management Science*).
- **Technological Advances:**
 - Computer simulations.
 - Integer programming, dynamic programming.

1940s

WWII applications

1957

First OR society
(INFORMS founded)

1970s

Spread to
business/industry

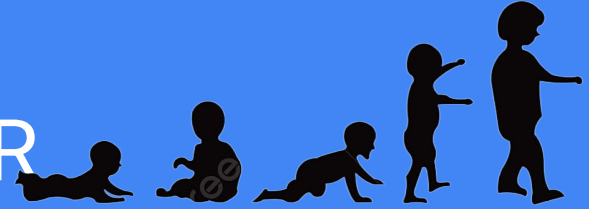
2000s

AI/OR integration

→ Modern Era (1990s–Present)

- **Computational Revolution:**
 - Big Data analytics.
 - Machine Learning + OR (e.g., Supply chain optimization).
- **Applications:** Healthcare, E-commerce

2. Stages of Development of OR



1. Problem Formulation: Define the real-world problem in clear, quantifiable terms.
2. Construction of a Mathematical Model: Represent the problem using equations or inequalities. Example: Linear Programming model for maximizing profit.
3. Model Solution: Use appropriate techniques (e.g., simplex method, dynamic programming) to find an optimal solution.
4. Model Validation: Check whether the model's output is consistent with the real system's behavior.
5. Implementation of Results: Apply the findings to make operational or strategic decisions.
6. Monitoring and Feedback: Evaluate the solution's performance over time and refine as needed.

3. Relationship: Manager and OR Specialist



Manager's Role:

- Provides domain-specific knowledge and constraints.
- Ensures alignment with business goals.

OR Specialist's Role:

- Constructs and solves the mathematical models.
- Offers optimal and feasible alternatives.



Importance of Collaboration: Successful OR application depends on continuous communication and trust between both parties.

Examples:

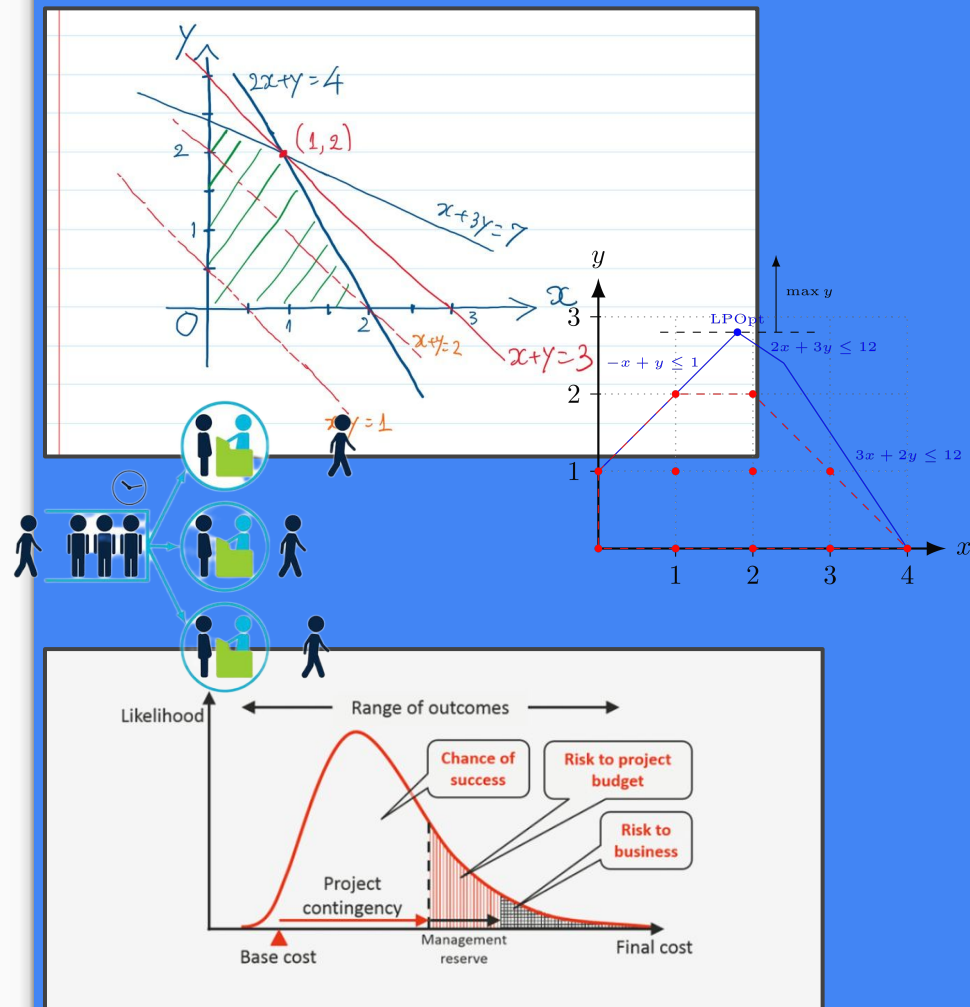
- A production manager explains resource limits; the OR specialist designs a model to optimize product mix.
- A delivery company faces high fuel costs due to inefficient routes. OR Solution: Use the Vehicle Routing Problem (VRP) to optimize delivery paths, minimizing distance/time.
- A hospital struggles with long patient wait times. Apply Queuing Theory to redesign staff schedules and reduce bottlenecks.
- A supermarket runs out of stock for popular items. Inventory Models (EOQ) predict optimal restocking levels.

- Energy
 - Problem: A power grid operator needs to meet demand at the lowest cost.
 - OR Solution: Linear Programming schedules power plants efficiently.
- Manufacturing
 - Problem: A factory has idle machines but misses deadlines.
 - OR Solution: Job-Shop Scheduling algorithms assign tasks to machines optimally.
- Agriculture
 - Problem: A farmer must decide crop mix for
 - OR Solution: Linear Programming models optimize
- Disaster Response
 - Problem: Relief supplies must reach disaster zones quickly amid road damage.
 - OR Solution: ...?
- Sports
 - Problem: ??
 - OR Solution: ??
- Telecommunications ...??



4. OR Tools & Techniques

- Mathematical Programming:
 - Linear Programming (LP)
 - Integer Programming (IP)
 - Nonlinear Programming (NLP)
- Simulation:
 - Monte Carlo simulation to analyze probabilistic systems.
- Queuing Theory:
 - Used to model waiting lines in banks, hospitals
- Inventory Models:
 - EOQ (Economic Order Quantity), Safety Stock calculations.
- Network Models:
 - Shortest path (Dijkstra's), Maximum flow (Ford-Fulkerson).
- Decision Analysis:
 - Decision trees, Utility theory.
- Game Theory:
 - Analyzing strategic interactions among rational agents.



“OR is the war-born science of efficiency.”



— Philip McCord Morse (Founder of MIT OR Center)

5. Limitations of Operations Research

1. Quantitative Nature: Difficult to model qualitative or emotional aspects (e.g., human behavior, ethics).
2. Data Dependency: Inaccurate or incomplete data can lead to misleading results.
3. Model Assumptions: Many models rely on assumptions like linearity, certainty, and stationarity which may not hold in real-life scenarios.

4. Complexity: Advanced models may require computational resources and skills not readily available.

5. Resistance to Change: Organizations may resist implementing OR solutions due to cultural or structural inertia.

A factory may not implement an optimal scheduling model due to union constraints or lack of skilled personnel.

Assignments

Can you TRY...?:

Take a scenario (e.g., a factory with two machines and three products).

- Formulate the linear programming model for maximizing profit
- Define decision variables, objective function, and constraints
- Solve graphically or using Excel Solver (for small instances)

1. Create a detailed visual timeline (hand-drawn or digital) showing the evolution of Operations Research from the 1940s to the present. Include key events like:
 - a. WWII military applications
 - b. Founding of INFORMS
 - c. OR in business and industry
 - d. AI/OR integration
 - e. Include names of major contributors (e.g., Patrick Blackett).

2. Choose a real-world application of OR (e.g., airline scheduling, supply chain optimization, hospital resource allocation).
 - Prepare a report (1000–1500 words) discussing:
 - The problem definition
 - OR technique(s) used
 - Benefits observed
 - Limitations faced

1. **Critically analyze** how the collaboration between managers and OR specialists influences the effectiveness of OR models. Provide real or hypothetical examples.
2. Compare and contrast the usefulness of **Simulation** and **Linear Programming** techniques in uncertain environments.
3. Why is **model validation** considered a critical step in the OR process? What are the possible consequences of skipping this step?
4. Discuss the **ethical implications** of using OR in fields like military operations or employee scheduling.
5. Evaluate the **applicability of classical OR techniques in today's data-driven and AI-integrated systems**. What adaptations are needed?



Thanks!

